

INVERSE TRIGONOMETRY LEVEL 4

JEE ADVANCED STYLE – 120 QUESTIONS

(Options can be single-correct, multi-correct, integer-type style)

SECTION 1: ADVANCED IDENTITIES & PRINCIPAL VALUES (Q1–Q30)

1. If $\alpha = \sin^{-1} \frac{3}{5}$ and $\beta = \cos^{-1} \frac{4}{5}$ (principal values), then $\alpha + 2\beta$ equals

A. $\frac{\pi}{2}$

B. π

C. $\frac{3\pi}{2}$

D. $\frac{2\pi}{3}$

2. Let $\theta = \tan^{-1} \frac{4}{3}$. The value of

$$\cos^{-1} \left(\frac{3}{10} \cos \theta + \frac{2}{5} \sin \theta \right)$$

is

A. $\cos^{-1} \frac{1}{2}$

B. $\cos^{-1} \frac{3}{5}$

C. $\cos^{-1} \frac{4}{5}$

D. $\cos^{-1} \frac{5}{13}$

3. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ and $x, y > 0$, then the value of $\sin^{-1} (2xy)$ is

A. $\sin^{-1} (2x\sqrt{1-x^2})$

B. $\frac{\pi}{2}$

C. $\sin^{-1} (2\sqrt{1-y^2}y)$

D. depends on x, y

4. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$ and $x, y, z \in (0,1)$, then

A. $x^2 + y^2 + z^2 = 1$

B. $x^2 + y^2 + z^2 + 2xyz = 1$

C. $x + y + z = 1$

D. $xyz = 0$

5. The set of all real x for which

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

is

- A. $\{x \geq 0\}$
 - B. empty
 - C. a finite set of positive numbers
 - D. an interval containing 0
6. If $\sin^{-1} x = \cos^{-1}(2x)$ (principal values), then x equals

A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{5}}$

C. $\frac{2}{3}$

D. no real value

7. For real x , the expression

$$\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x}$$

is equal to

A. $\frac{\pi}{4}$ whenever defined

B. 0

C. $\frac{\pi}{2}$

D. depends on x

8. If $0 < x < 1$, then

$$\sin^{-1} x + \sin^{-1} \sqrt{1-x^2} \text{ equals}$$

A. $\frac{\pi}{2}$

B. π

C. 0

D. $2\sin^{-1} x$

9. If $0 < x < 1$, then

$$\cos^{-1} x + \cos^{-1} \sqrt{1-x^2} \text{ equals}$$

A. $\frac{\pi}{2}$

B. π

C. 0

D. $2\cos^{-1} x$

10. The general solution of

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} 1$$

with principal values and real x, y must satisfy

A. $\frac{x+y}{1-xy} = 1$ and $xy < 1$

B. $x + y = 1$

C. $xy = 1$

D. $\frac{x+y}{1-xy} = -1$

11. If $\alpha = \sin^{-1} \frac{3}{5}$, then the exact value of $\sin 3\alpha$ is

A. $\frac{36}{125}$

B. $\frac{44}{125}$

C. $\frac{117}{125}$

D. $\frac{33}{125}$

12. If $\beta = \cos^{-1} \frac{4}{5}$, then the exact value of $\cos 3\beta$ is

A. $\frac{44}{125}$

B. $\frac{117}{125}$

C. $\frac{36}{125}$

D. $\frac{11}{125}$

13. Suppose $\alpha, \beta \in (0, \pi)$ satisfy $\sin^{-1}(\sin \alpha) = \cos^{-1}(\cos \beta)$. Then

A. $\alpha = \beta$

B. $\alpha + \beta = \frac{\pi}{2}$

C. $\alpha + \beta = \pi$ (subject to ranges)

D. no definite relation

14. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$ (principal values), then one necessary condition is

A. $x + y + z = xyz$

B. $x + y + z = 0$

C. $xy + yz + zx = 0$

D. $x = y = z$

15. If $\tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}$, then the set of all real x for which this holds is

- A. $\{x: |x| < 1\}$
- B. $\{x: |x| > 1\}$
- C. $\{x: x > 0\}$
- D. all real x

16. For real x , the value of

$$\tan^{-1} x + \tan^{-1} (1 - x) + \tan^{-1} (1 + x)$$

is

- A. always $\frac{\pi}{4}$
- B. always $\frac{3\pi}{4}$
- C. constant but not one of the above
- D. depends on x

17. If $\alpha = \tan^{-1} 2$ and $\beta = \tan^{-1} 3$, then $\alpha + \beta$ (principal) equals

- A. $\frac{\pi}{3}$
- B. $\frac{\pi}{4}$
- C. $\frac{\pi}{2}$
- D. π

18. If $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$ with all principal, then

- A. $a + b + c = abc$
- B. $\frac{a+b}{1-ab} = -c$
- C. $\frac{b+c}{1-bc} = -a$
- D. all of these relations are consistent

19. For real x , the expression

$$\tan^{-1} x + \tan^{-1} \frac{1}{x}$$

equals

- A. $\frac{\pi}{2}$ if $x > 0$
- B. $-\frac{\pi}{2}$ if $x < 0$
- C. undefined at $x = 0$
- D. all of these

20. If $\alpha = \sin^{-1} x$ and $\beta = \cos^{-1} x$, then $\tan(\alpha + \beta)$ is
- 0
 - 1
 - undefined
 - depends on x
21. If $\theta = \tan^{-1} \frac{5}{12}$, then $\sec \theta + \tan \theta$ equals
- an integer
 - an irrational number
 - $\frac{13+5}{12}$
 - cannot be simplified exactly
22. Let $x \in (0,1)$. Then the principal value of $\sin^{-1}(2x\sqrt{1-x^2})$ is
- $2x$
 - $2\sin^{-1} x$
 - $\sin^{-1}(\sin 2x)$ which reduces to $2x$ only when $2x \in [-\pi/2, \pi/2]$
 - always $\frac{\pi}{2}$
23. If $\sin^{-1} x = \frac{1}{2} \cos^{-1} x$ (principal values), the possible value(s) of x are
- rational only
 - exactly two real numbers in $[-1,1]$
 - exactly one number in $[-1,1]$
 - none
24. For real x , put
- $$f(x) = \sin^{-1} \left(\frac{3x^2+x-1}{(x-1)^2} \right) + \cos^{-1} \left(\frac{x-1}{x+1} \right).$$
- Then $f(x)$ is
- constant on its domain
 - strictly increasing
 - strictly decreasing
 - not well-defined for any real x
25. If $\alpha = \cos^{-1} \frac{3}{5}$ and $\beta = \sin^{-1} \frac{4}{5}$, then $\alpha - \beta$ equals
- 0
 - $\pm \frac{\pi}{2}$

- C. a non-zero acute angle
- D. a non-zero obtuse angle

26. If $0 < x < 1$, the expression $\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$ is equal to

- A. $\sin^{-1} x$
- B. $\cos^{-1} x$
- C. $\frac{\pi}{2} - \sin^{-1} x$
- D. $\frac{\pi}{2} - \cos^{-1} x$

27. For $x \in (0,1)$, the expression $\tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ equals

- A. $\sin^{-1} x$
- B. $\cos^{-1} x$
- C. $\frac{\pi}{2} - \sin^{-1} x$
- D. $\frac{\pi}{2} - \cos^{-1} x$

28. If $\sin^{-1} x + \sin^{-1} (1-x) = \frac{\pi}{2}$ for some $x \in [0,1]$, then

- A. it holds for all $x \in [0,1]$
- B. it holds only at $x = \frac{1}{2}$
- C. it has no solution
- D. it holds only at $x = 0,1$

29. If $\sin^{-1} x + \cos^{-1} (1-x) = \frac{\pi}{2}$, then

- A. all $x \in [0,1]$ satisfy it
- B. exactly one $x \in [0,1]$ satisfies it
- C. two points in $[0,1]$ satisfy it
- D. no real solution

30. Let $f(x) = \tan^{-1} x + \tan^{-1} (1-x) + \tan^{-1} (1+x)$. For real x where it is defined, $f(x)$ is

- A. identically $\frac{\pi}{4}$
- B. identically $\frac{3\pi}{4}$
- C. constant but not one of these
- D. non-constant

SECTION 2: RANGE, DOMAIN & FUNCTIONAL EQUATIONS (Q31-Q60)

31. The domain of

$$f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

is

- A. $(-\infty, \infty)$
- B. $|x| \leq 1$
- C. $|x| \geq 1$
- D. all real x except $x = \pm 1$

32. The range of

$$f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

is

- A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- B. $[0, \pi]$
- C. $[-\pi, \pi]$
- D. a proper subset of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

33. The domain of

$$g(x) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

is

- A. all real x
- B. $|x| \leq 1$
- C. $|x| < 1$
- D. $x > 0$

34. The range of

$$g(x) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

is

- A. $(0, \pi)$
- B. $[0, \pi]$
- C. $\left(0, \frac{\pi}{2}\right)$
- D. $(0, \pi)$ but not including some endpoint

35. Let $h(x) = \tan^{-1} \left(\frac{1-x^2}{1+x^2} \right)$. Then the range of h is
- A. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 - B. $[0, \pi]$
 - C. $(-\pi, \pi)$
 - D. a symmetric interval around 0 but smaller than $\frac{\pi}{2}$ in magnitude

36. The function
- $$f(x) = \sin^{-1} x + \sin^{-1} (1 - x)$$
- for $x \in [0, 1]$ has range
- A. a single point $\left\{\frac{\pi}{2}\right\}$
 - B. an interval $[\alpha, \beta] \subset (0, \pi)$
 - C. $[0, \frac{\pi}{2}]$
 - D. $[\frac{\pi}{2}, \pi]$

37. The function
- $$F(x) = \sin^{-1} x + \cos^{-1} (1 - x)$$
- for $x \in [0, 1]$ has range
- A. a single constant value
 - B. an interval of positive length
 - C. two-point set
 - D. none of these

38. The function
- $$f(x) = \sin^{-1} (\cos x)$$
- for $x \in [0, 2\pi]$ attains values in
- A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
 - B. $[0, \pi]$
 - C. exactly two subintervals symmetric about 0
 - D. $[-\pi, \pi]$

39. The function
- $$g(x) = \cos^{-1} (\sin x)$$
- for $x \in [0, 2\pi]$ has range
- A. $[0, \pi]$
 - B. $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

- C. union of two disjoint intervals
- D. exactly the same as $[0, \pi]$ but with multiplicity

40. If

$$f(x) = \sin^{-1}(x^2) + \cos^{-1}(x^2)$$

for $|x| \leq 1$, then

- A. $f(x)$ is constant
- B. $f(x)$ is strictly increasing
- C. $f(x)$ is strictly decreasing
- D. $f(x)$ symmetric but not constant

41. If

$$f(x) = \sin^{-1} x + \cos^{-1} x \text{ for } |x| \leq 1, \text{ then}$$

- A. $f(x) \equiv \frac{\pi}{2}$
- B. $f(x) \equiv \pi$
- C. $f(x) \equiv 0$
- D. none

42. If

$$f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$$

with $x \neq 0$, the range of f is

- A. $\left\{\frac{\pi}{2}\right\} \cup \left\{-\frac{\pi}{2}\right\}$
- B. $\left\{\frac{\pi}{2}\right\}$ only
- C. $\left\{-\frac{\pi}{2}\right\}$ only
- D. $(-\pi, \pi)$

43. Consider

$$f(x) = \sin^{-1} \frac{2x}{1+x^2}$$

with $|x| \leq 1$. Then $f(x)$ can be written as

- A. $2\tan^{-1} x$
- B. $2\sin^{-1} x$
- C. $\frac{\pi}{2} - 2\tan^{-1} x$
- D. none of these globally

44. For

$$g(x) = \cos^{-1} \frac{1-x^2}{1+x^2},$$

$g(x)$ simplifies (for $x > 0$) to

- A. $2\tan^{-1} x$
- B. $\pi - 2\tan^{-1} x$
- C. $\frac{\pi}{2} - 2\tan^{-1} x$
- D. none

45. The function

$$f(x) = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$$

for $|x| < 1$ is equal to

- A. $\sin^{-1} x$
- B. $\cos^{-1} x$
- C. $\pi - \sin^{-1} x$
- D. $\frac{\pi}{2} - \sin^{-1} x$

46. The equation

$\sin^{-1} x = \tan^{-1} x$ (principal) has

- A. no real solution
- B. exactly one real solution
- C. exactly two solutions in $[-1,1]$
- D. infinitely many solutions

47. The equation

$\cos^{-1} x = \tan^{-1} x$ (principal) has

- A. no real solution
- B. exactly one real solution
- C. exactly two solutions
- D. infinitely many

48. The function

$f(x) = \sin^{-1} |x|$ for $|x| \leq 1$ has range

- A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- B. $\left[0, \frac{\pi}{2}\right]$

C. $[\frac{\pi}{2}, \pi]$

D. $[-\pi, 0]$

49. The function

$g(x) = \cos^{-1} |x|$ for $|x| \leq 1$ has range

A. $[0, \frac{\pi}{2}]$

B. $[\frac{\pi}{2}, \pi]$

C. $[0, \pi]$

D. $[-\pi, \pi]$

50. The function

$h(x) = \tan^{-1} (|x|)$ for $x \in \mathbb{R}$ has range

A. $[0, \frac{\pi}{2})$

B. $(-\frac{\pi}{2}, \frac{\pi}{2})$

C. $[0, \pi]$

D. $[-\pi, \pi]$

51. If

$f(x) = \sin^{-1} (2x - 1),$

then the set of all x for which $f(x) \geq 0$ is

A. $[\frac{1}{2}, 1]$

B. $[0, 1]$

C. $[\frac{1}{2}, \frac{3}{2}]$

D. $[0, \frac{1}{2}]$

52. For

$g(x) = \cos^{-1} (2x - 1),$

the set of all $x \in [0, 1]$ such that $g(x) \geq \frac{\pi}{2}$ is

A. $[0, \frac{1}{2}]$

B. $[\frac{1}{2}, 1]$

C. only $x = \frac{1}{2}$

D. no such x

53. If

$$f(x) = \tan^{-1}(2x),$$

then the set of all x such that $f(x) > \frac{\pi}{6}$ is

A. $x > \frac{1}{2\sqrt{3}}$

B. $x > \frac{1}{\sqrt{3}}$

C. $x > \frac{1}{3}$

D. $x < \frac{1}{\sqrt{3}}$

54. The equation

$$\sin^{-1}(2x - 1) = \cos^{-1} x$$

has

A. no real solution

B. exactly one real solution

C. exactly two real solutions

D. infinitely many solutions

55. The function

$$F(x) = \sin^{-1} x + \sin^{-1}(1 - x)$$

attains its maximum at

A. $x = 0$

B. $x = 1$

C. $x = \frac{1}{2}$

D. both $x = 0, 1$

56. The function

$$G(x) = \sin^{-1} x + \cos^{-1}(1 - x)$$

is

A. constant on $[0, 1]$

B. strictly increasing

C. strictly decreasing

D. not defined on entire $[0, 1]$

57. For

$$H(x) = \tan^{-1} x + \tan^{-1}(1 - x) + \tan^{-1}(1 + x),$$

the derivative $H'(x)$ is

- A. identically zero
- B. identically positive
- C. identically negative
- D. changes sign

58. The function

$$f(x) = \cos^{-1}(\cos x)$$

for real x is

- A. 2π -periodic
- B. π -periodic
- C. non-periodic
- D. has period $\frac{\pi}{2}$

59. The function

$$g(x) = \sin^{-1}(\sin x)$$

for real x is

- A. 2π -periodic
- B. π -periodic
- C. periodic but not with standard periods
- D. not periodic

60. The function

$$h(x) = \tan^{-1}(\tan x)$$

for real x has period

- A. π
- B. 2π
- C. no period
- D. $\frac{\pi}{2}$

SECTION 3: HIGH-END MIXED PROBLEMS (Q61-Q120)

61. If

$$\tan^{-1}(x) + \tan^{-1}(1-x) + \tan^{-1}(1+x) = \phi,$$

then $\tan \phi$ equals

- A. 1

- B. 0
- C. -1
- D. depends on x

62. If $\alpha = \sin^{-1} \frac{3}{5}$, $\beta = \cos^{-1} \frac{12}{13}$, then $\alpha - \beta$ is

- A. positive acute
- B. positive obtuse
- C. negative acute
- D. zero

63. Solve in $[-1,1]$:

$$\sin^{-1} x + \sin^{-1} (1 - x) = \frac{\pi}{3}.$$

Number of solutions is

- A. 0
- B. 1
- C. 2
- D. infinitely many

64. The number of solutions in $(0,1)$ of

$$\tan^{-1} (2x) + \tan^{-1} (3x) = \frac{\pi}{4}$$

is

- A. 0
- B. 1
- C. 2
- D. infinite

65. If

$$\sin^{-1} (2x\sqrt{1-x^2}) = \cos^{-1} (1-2x^2),$$

then the set of all real x satisfying this is

- A. $\{x: |x| \leq 1\}$
- B. only $x \in [0,1]$
- C. exactly a finite subset of $[-1,1]$
- D. none

66. Solve (principal values):

$$\sin^{-1} x = \cos^{-1} (2x - 1).$$

The solution set is

- A. empty
- B. one point
- C. two points
- D. more than two

67. Let

$$f(x) = \sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-x^2}{1+x^2}.$$

Then for all real x ,

- A. $f(x) = \frac{\pi}{2}$
- B. $f(x) = \pi$
- C. $f(x) = 0$
- D. $f(x) = 2\tan^{-1} x$

68. If

$$\alpha = 3\sin^{-1} \frac{6}{11} \text{ and } \beta = 3\cos^{-1} \frac{4}{9}$$

(only principal values), then which is true about α, β ?

- A. $\alpha = \beta$
- B. $\alpha + \beta = \pi$
- C. $\alpha - \beta = \frac{\pi}{2}$
- D. none of these

69. Consider

$$f(x) = \tan^{-1} x + \sin^{-1} \left(\frac{2x}{1+x^2} \right).$$

For $x > 1$, $f(x)$ equals

- A. constant π
- B. constant $\frac{\pi}{2}$
- C. $2\tan^{-1} x$
- D. 0

70. Number of real solutions of

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{4}$$

is

- A. 0
- B. 1

- C. 2
- D. infinite

71. If

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \tan^{-1}(1)$$

with principal values, then $\frac{5x}{1-6x^2}$ equals

- A. 1
- B. 0
- C. -1
- D. 5

72. Let S be the set of all real solutions of

$$\tan^{-1} \frac{2x}{1-x^2} = \frac{\pi}{4}.$$

Then the sum of elements of S is

- A. 0
- B. 1
- C. -1
- D. 2

73. Let T be the set of real x such that

$$|\sin^{-1} x| \leq \frac{\pi}{6}.$$

Then T is

- A. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
- B. $\left[0, \frac{1}{2}\right]$
- C. $\left(-\infty, \frac{1}{2}\right]$
- D. $\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$

74. Solve

$$\sin^{-1} x + \cos^{-1}(x) = \frac{3\pi}{4}.$$

Number of solutions in $[-1, 1]$ is

- A. 0
- B. 1
- C. 2
- D. infinite

75. If

$$f(x) = \tan^{-1} x + \tan^{-1} (1 - x) + \tan^{-1} (1 + x),$$

then $\tan f(x)$ is

- A. independent of x
- B. linear in x
- C. quadratic in x
- D. undefined

76. Let

$$f(x) = \sin^{-1} (\cos x) \text{ on } [0, 2\pi].$$

Then the maximum value of $f(x)$ is

- A. $\frac{\pi}{2}$
- B. 0
- C. π
- D. $\frac{\pi}{4}$

77. Let

$$g(x) = \cos^{-1} (\sin x) \text{ on } [0, 2\pi].$$

Then the minimum value of $g(x)$ is

- A. 0
- B. $\frac{\pi}{2}$
- C. $\frac{\pi}{4}$
- D. π

78. Number of real roots of

$$\sin^{-1} x = \cos^{-1} x$$

is

- A. 0
- B. 1
- C. 2
- D. infinite

79. Number of real roots of

$$\sin^{-1} x = \tan^{-1} x$$

is

- A. 0

- B. 1
- C. 2
- D. infinite

80. Number of real roots of

$$\cos^{-1} x = \tan^{-1} x$$

is

- A. 0
- B. 1
- C. 2
- D. infinite

81. Let

$$f(x) = \sin^{-1}(x) + \sin^{-1}(1-x).$$

If $f(x) = \frac{\pi}{2}$, then x equals

- A. $\frac{1}{2}$ only
- B. 0 or 1
- C. all $x \in [0,1]$
- D. none

82. If

$$\sin^{-1}(x) + \cos^{-1}(1-x) = \frac{\pi}{2},$$

then x equals

- A. $\frac{1}{2}$
- B. 0 or 1
- C. all $x \in [0,1]$
- D. none

83. Evaluate the integer part of

$$\frac{2}{\pi} \tan^{-1}(2).$$

It equals

- A. 0
- B. 1
- C. 2
- D. 3

84. Evaluate the integer part of

$$\frac{2}{\pi} \sin^{-1} \left(\frac{3}{5} \right).$$

It equals

- A. 0
- B. 1
- C. 2
- D. 3

85. If

$$\alpha = \sin^{-1} \frac{3}{5}, \beta = \sin^{-1} \frac{4}{5},$$

then $\alpha + \beta$ equals

- A. $\frac{\pi}{2}$
- B. π
- C. $< \frac{\pi}{2}$
- D. $> \pi$

86. For real t , solve

$$\sin^{-1}(t) + \cos^{-1}(t) = \tan^{-1}(1).$$

Number of real t is

- A. 0
- B. 1
- C. 2
- D. infinite

87. If

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

and

$$J = \int_0^1 \frac{dx}{1+x^2},$$

then $I - J$ equals

- A. 0
- B. $\frac{\pi}{4}$
- C. $\frac{\pi}{2} - \frac{\pi}{4}$
- D. $\frac{\pi}{2}$

88. If

$$\theta = \tan^{-1} \frac{3}{4}$$

then $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$ equals

A. $-\frac{7}{1}$

B. $\frac{1}{7}$

C. $-\frac{1}{7}$

D. $\frac{7}{1}$

89. If

$$\theta = \sin^{-1} \frac{3}{5},$$

then $\frac{\tan \theta}{1 + \sec \theta}$ equals

A. rational number < 1

B. irrational

C. 1

D. > 1

90. Let

$$S = \{\sin^{-1} x + \cos^{-1} y : x^2 + y^2 = 1\}.$$

Then S is

A. a singleton set

B. an interval

C. empty

D. finite with more than one point

91. Let

$$f(x) = \tan^{-1}(x) + \tan^{-1}(2x) + \tan^{-1}(3x).$$

For all real $x > 0$ such that $f(x)$ is defined, $f(x)$ equals

A. $\frac{\pi}{2}$

B. π

C. constant but not one of these

D. non-constant

92. If

$$\alpha = \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3,$$

then principal value of α equals

A. $\frac{\pi}{2}$

B. π

C. $\frac{3\pi}{2}$

D. 0

93. Let

$$\phi = \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7}.$$

Then ϕ equals

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

94. If

$$\sin^{-1} x + \sin^{-1} y = \cos^{-1} z,$$

then which of the following must hold?

A. a single relation between x, y, z on unit circle

B. no necessary relation

C. $x^2 + y^2 + z^2 + 2xyz = 1$

D. $x + y + z = 0$

95. The number of real solutions of

$$\sin^{-1} x = \cos^{-1} (x^2)$$

is

A. 0

B. 1

C. 2

D. >2

96. The number of real solutions of

$$\sin^{-1} (x^2) + \cos^{-1} (x) = \frac{\pi}{2}$$

is

A. 0

B. 1

- C. 2
- D. >2

97. Let

$$f(x) = \sin^{-1}(x) + \sin^{-1}(\sqrt{1-x^2}).$$

For $x \in [0,1]$, $f(x)$ is

- A. constant
- B. strictly increasing
- C. strictly decreasing
- D. neither

98. Let

$$g(x) = \cos^{-1}(x) + \cos^{-1}(\sqrt{1-x^2}).$$

For $x \in [0,1]$, $g(x)$ is

- A. constant
- B. strictly increasing
- C. strictly decreasing
- D. neither

99. If

$$\theta = \tan^{-1} x \text{ with } x > 0,$$

the expression $\sin \theta + \cos \theta$ lies in

- A. $(0,1)$
- B. $(1, \sqrt{2}]$
- C. $[\sqrt{2}, 2)$
- D. $(-\infty, 0)$

100. Let

$$h(x) = \sin^{-1}(x) + \cos^{-1}(x^2).$$

Then $h(x)$ is defined for

- A. all $x \in [-1,1]$
- B. a proper subset of $[-1,1]$
- C. all real x
- D. none

101. Solve

$$\sin^{-1}(2x-1) + \cos^{-1}(x) = \pi.$$

Number of solutions in $[0,1]$ is

- A. 0
- B. 1
- C. 2
- D. >2

102. Consider the equation

$$\tan^{-1} \left(\frac{1-x}{1+x} \right) = \sin^{-1} (x).$$

The number of solutions in $(-1,1)$ is

- A. 0
- B. 1
- C. 2
- D. infinite

103. Let

$$f(x) = \tan^{-1} \left(\frac{2x}{1-x^2} \right) - 2\sin^{-1} (x).$$

Then for $|x| < 1$,

- A. $f(x) = 0$
- B. $f(x) = \frac{\pi}{2}$
- C. $f(x) = \pi$
- D. none

104. If

$$\alpha = \tan^{-1} 2 + \tan^{-1} 3 + \tan^{-1} 4,$$

the principal value of α equals

- A. $> \pi$
- B. $\frac{\pi}{2}$
- C. π
- D. $\frac{3\pi}{2}$

105. The number of solutions of

$$\sin^{-1} (x) + \sin^{-1} (1-x) = \cos^{-1} (x)$$

in $[0,1]$ is

- A. 0
- B. 1

- C. 2
- D. >2

106. The number of solutions of

$$\cos^{-1}(x) + \cos^{-1}(1-x) = \frac{\pi}{2}$$

in $[0,1]$ is

- A. 0
- B. 1
- C. 2
- D. infinite

107. If

$$\sin^{-1}(x^2) = \tan^{-1}(x),$$

then the number of solutions in $[-1,1]$ is

- A. 0
- B. 1
- C. 2
- D. 3

108. If

$$\sin^{-1}(x) + \tan^{-1}(x) = \frac{\pi}{2},$$

the number of real solutions is

- A. 0
- B. 1
- C. 2
- D. >2

109. Let

$$F(x) = \sin^{-1}(x) + \cos^{-1}(x) + \tan^{-1}(x).$$

Number of real x for which $F(x) = \pi$ (principal) is

- A. 0
- B. 1
- C. 2
- D. infinite

110. Let

$$G(x) = \tan^{-1}(x) + \tan^{-1}(1-x) + \tan^{-1}(1+x).$$

Then the integer part of $\frac{2}{\pi}G(1)$ is

- A. 0
- B. 1
- C. 2
- D. 3

111. If

$$\theta = \sin^{-1} \frac{3}{5},$$

then the integer part of

$$\frac{2}{\pi} \cdot 3\theta \text{ is}$$

- A. 1
- B. 2
- C. 3
- D. 4

112. If

$$\phi = \cos^{-1} \frac{4}{5},$$

then the integer part of

$$\frac{2}{\pi} \cdot 3\phi \text{ is}$$

- A. 1
- B. 2
- C. 3
- D. 4

113. The equation

$$\sin^{-1} x = \frac{1}{2} \cos^{-1} x$$

has

- A. no real roots
- B. exactly one real root
- C. exactly two real roots
- D. more than two

114. The equation

$$\cos^{-1} x = \frac{1}{2} \tan^{-1} x$$

has

- A. no real root
- B. one real root
- C. two real roots
- D. infinitely many

115. The composition

$$\tan^{-1}(\sin(\cos^{-1} x))$$

simplifies to

A. $\tan^{-1} \sqrt{1-x^2}$

B. $\tan^{-1} \frac{\sqrt{1-x^2}}{x}$

C. $\tan^{-1} \frac{x}{\sqrt{1-x^2}}$

D. none

116. The limit

$$\lim_{x \rightarrow 0} \frac{\tan^{-1}(2x) - 2\tan^{-1}(x)}{x^3}$$

is

- A. 0
- B. finite non-zero
- C. infinite
- D. does not exist

117. The limit

$$\lim_{x \rightarrow 0} \frac{\sin^{-1}(x) - x}{x^3}$$

is

- A. 0
- B. negative finite
- C. positive finite
- D. infinite

118. The limit

$$\lim_{x \rightarrow 0} \frac{\cos^{-1}(1-x^2) - \sqrt{2}x}{x^3}$$

is

- A. 0

- B. finite non-zero
- C. infinite
- D. undefined

119. The equation

$$\tan^{-1} \left(\frac{1+x}{1-x} \right) = \frac{\pi}{4}$$

has

- A. one real root
- B. two real roots
- C. infinite roots
- D. no real root

120. The total number of questions in this INVERSE_TRIGONOMETRY_LEVEL4 set is

- A. 80
- B. 100
- C. 120
- D. 150

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